Project Design Phase-I

Solution Architecture

Date	04 March 2026
Team ID	EL2026TMID4684
Project Name	Insurance Fraud Detection System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram — Insurance Fraud Detection System:

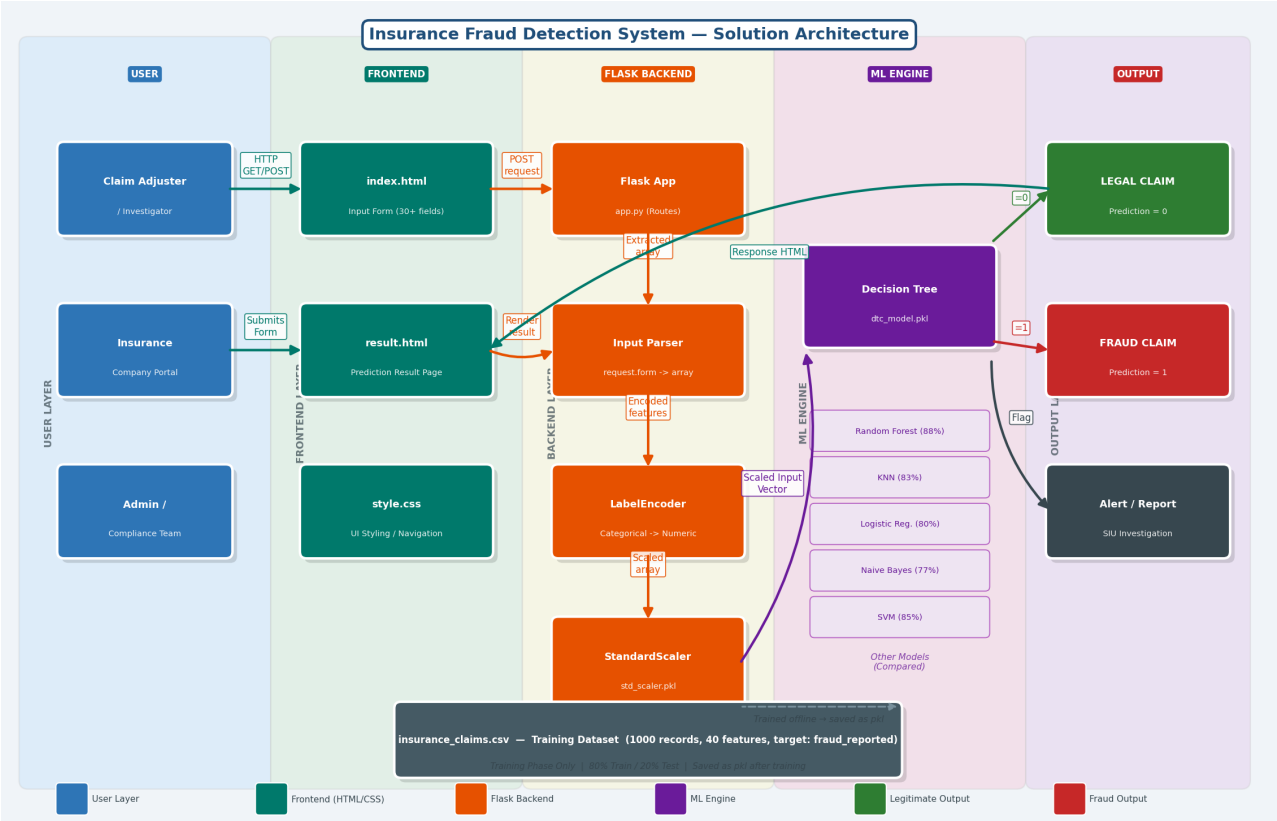

Figure 1: Solution Architecture and Data Flow of the Insurance Fraud Detection System — showing User Layer, Frontend (HTML/CSS), Flask Backend, ML Engine (Decision Tree Classifier), and Output Layer with Fraud / Legal prediction results.

Architecture — Layer-by-Layer Description:

Layer	Components	Technology	Role
User Layer	Claim Adjuster, Insurance Portal, Compliance Team	Web Browser	Submits claim details via web form; receives prediction result
Frontend Layer	index.html, result.html, style.css	HTML5 / CSS3	Renders 30+ input fields; displays Legal/Fraud prediction to user
Flask Backend	app.py — Routes: GET /, POST /predict Input Parser	Python / Flask 2.x	Receives POST data, parses form values, coordinates preprocessing & inference
Preprocessing	LabelEncoder, StandardScaler (std_scaler.pkl)	Scikit-learn / Pickle	Encodes categorical features; normalizes numeric array before model input
ML Engine	Decision Tree Classifier (dtc_model.pkl)	Scikit-learn / Pickle	Predicts: 0 = Legal Insurance Claim | 1 = Fraud Insurance Claim
Output Layer	Legal / Fraud Result Page SIU Alert / Report	Flask render_template	Displays colored prediction result; routes fraud cases to SIU investigation
Training Dataset	insurance_claims.csv (1000 records, 40 features)	Pandas / NumPy	Used offline to train & compare 6 ML models; saved as pkl after training

Reference: https://scikit-learn.org/stable/ | https://flask.palletsprojects.com/ | https://aws.amazon.com/architecture/